Lecture Notes on

Stars and their Life Cycles

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These notes summarize the third Astrophysics Club meeting, where we explored how stars form, evolve, and die.

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1 Why Stars Matter

Stars are central to astrophysics because:

- They generate nearly all visible light in the universe.
- They create heavier elements through nuclear fusion.
- Their evolution shapes galaxies.
- Their remnants (white dwarfs, neutron stars, black holes) drive extreme physics.

2 Star Formation

Star formation occurs in three major stages.

2.1 1. Molecular Cloud Collapse

- Stars form in cold molecular clouds (nebulae).
- These clouds are primarily hydrogen gas.
- Gravity competes against internal gas pressure.

Collapse occurs when the cloud exceeds the **Jeans Mass**:

$$M_J \propto \left(\frac{T^{3/2}}{\rho^{1/2}} \right).$$

Interpretation:

- Higher temperature $\rightarrow$ harder to collapse.
- Higher density $\rightarrow$ easier to collapse.

Once gravity dominates, the cloud contracts.

2.2 2. Protostar Formation

During collapse:

- Gravitational potential energy converts to thermal energy.
- Core temperature rises.
- Angular momentum conservation causes rotation.
- A protoplanetary disk forms.

This disk later forms planets and asteroids.

2.3 3. Fusion Ignition

When core temperature reaches:

$$T \approx 10^7 \text{ K},$$

hydrogen fusion begins.
Main reaction (simplified):

$$4\,{}^1H \rightarrow {}^4He + \text{energy}.$$

This initiates:
Hydrostatic equilibrium:

- Gravity pulls inward.
- Thermal pressure from fusion pushes outward.
- Balance creates a stable star.

The star now enters the **main sequence**.

3 Stellar Evolution and Mass Dependence

A star's mass determines its fate.
Key factors:

- Core temperature
- Fusion limits
- Gravity vs pressure balance
- Degeneracy pressure

Fusion beyond iron does not release energy. Once iron forms, collapse becomes inevitable.

4 Low-Mass Stars ($< 8M_\odot$)

- Fuse hydrogen for billions of years.
- Expand into red giants.
- Outer layers shed (planetary nebula).
- Core remains as a **white dwarf**.

White dwarf properties:

- Supported by **electron degeneracy pressure**.
- Maximum mass: Chandrasekhar limit

$$M_{\max} \approx 1.4M_\odot.$$

Beyond this limit, collapse continues.

5 Intermediate-Mass Stars (8–$20M_\odot$)

- Fuse heavier elements successively.
- Iron core forms.
- Core collapses rapidly.
- Triggers a **core-collapse supernova**.
- Remnant becomes a **neutron star**.

Neutron star:

- Diameter 12 miles.
- Supported by neutron degeneracy pressure.
- Extremely dense (teaspoon ≈ billions of tons).

6 High-Mass Stars (> 20–$25M_\odot$)

- Similar fusion chain but much stronger gravity.
- Collapse cannot be stopped by neutron pressure.
- Forms a **black hole**.

Supernova explosion ejects heavy elements into space, enriching future generations of stars.

7 Life–Death Summary

Star life cycle:

- Molecular cloud → protostar → main sequence.
- Red giant / supergiant phase.
- Final remnant determined by mass:
  - White dwarf
  - Neutron star
  - Black hole

Mass is the dominant variable controlling fate.

8 Hertzsprung–Russell Diagram

The H–R diagram plots:

- Horizontal axis: surface temperature (decreasing left to right).
- Vertical axis: luminosity (increasing upward).

Key regions:

- **Main sequence**: hydrogen fusion.
- **Giants / supergiants**: expanded, luminous stars.
- **White dwarfs**: hot but faint remnants.

Physical meaning:

- Position reflects internal structure.
- Movement across diagram tracks stellar evolution.

9 Exotic Future: Iron Stars

Hypothetical:

- After $\sim 10^{1500}$ years, quantum tunneling may convert matter to iron.
- Iron stars would represent extremely late cosmic evolution.

This is far beyond current stellar timescales.